TOPIC 03 · FINANCIAL MANAGEMENT (I)

# Financial Management (I): Financial Statements and Financial Structure

Economic and financial structure, the income statement, the balance sheet, the sources of financing, the debt-to-equity ratio and the weighted average cost of capital.

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OFFICIAL MATERIAL `Topic 3 Financial Management (I).pdf`

`T3_Financial Statements_EXERCISES.pdf`

**Central idea:** every firm has two sides. The **economic structure** (assets: what is owned) is where the money is invested and it generates **profitability**; the **financial structure** (equity + liabilities: what is owed) is where the money comes from and it has a **cost**. The firm creates value when **profitability > cost**. Two statements describe the firm: the **income statement** (performance over a period: revenues − expenses = net income) and the **balance sheet** (position at a point in time: assets = liabilities + equity). The financial structure is the mix of **self-financing, external equity, long-term and short-term debt**, measured by the **debt-to-equity ratio** and priced by the **weighted average cost of capital**.

This note follows the official deck slide by slide. Every definition is taken from the slides; the worked numbers are computed here from the data printed on the slides. Citations such as [Slide 12] open the PDF on that slide. The step-by-step method to build these statements in the exercises is on the problems page.

**Contents of the topic [Slide 2]:** introduction: economic and financial structure; financial statements: the income statement and the balance sheet; the financial structure of the company.

## Part I — Introduction: economic and financial structure

### § 1 The finance area and its two decisions [Slide 3]

FINANCE AREA [SLIDE 3]

The **finance area** of a company refers to the **collection and management of the funds (capital)** used by the company.

The **finance manager** faces two important types of decisions while managing a company:

| Question | Decision |
|---|---|
| What to invest in and when? | **Investing decisions** |
| How to obtain the money to invest? | **Financing decisions** |

## § 2 Economic structure and financial structure [Slide 4 ]

| | Economic structure | Financial structure |
|---|---|---|
| Defined by | The **investments** carried out by the company | The **sources of financing** of the company |
| What it represents | The **use** of the financial resources in terms of **goods (assets) and rights** | **All the financial resources** used to carry out the investments (debt and liabilities) |
| Name | **ASSETS (what is owned)** | **LIABILITIES (what is owed)** |

The two structures are two views of the **same money**: the financial structure says where each euro came from, the economic structure says what it was spent on. That is why, in the balance sheet, both sides always add up to the same total (§8).

## § 3 The objective: maximize value [Slide 5 ]

The slide draws the balance as two columns: **ASSETS** on the left and **EQUITY + LIABILITIES** on the right. Assets generate **profitability**; equity and liabilities carry a **cost** (the arrows show that the profitability of the assets has to pay for the cost of the funds).

$$\text{OBJECTIVE: MAXIMIZE VALUE} \iff \text{PROFITABILITY} > \text{COST}$$

## § 4 What the company's decisions focus on [Slide 6 ]

**Financing decisions:**

- they refer to the **choice of equity or debt**;
- the financial resources that a company may use can be **internal equity** (self-financing: retained earnings), **external equity** or **debt**;
- financial decisions involve **decisions about the capital structure** and **decisions about the dividends**.

**Investing decisions:**

- they consist in the **allocation of capital** to those investment projects from which we expect future profits;
- companies will invest in those projects that will **create value** for them (**profits > costs**).

# Part II — Financial statements

## § 5 The income statement: basic format [Slide 7 ]

- The **first statement** prepared is the income statement.
- The income statement reports the **business performance for a specific period** (a quarter, a semester, a year).
- A simple format is:

$$\text{Revenues} - \text{Expenses} = \text{Net income}$$

## § 6 Revenues and expenses [Slide 8 ]

REVENUES AND EXPENSES [SLIDE 8]

**Revenues** are earned for the sale of goods or services. Revenues occur **when the sale is made**; the payment **may or may not have been received**. Examples: sales, service revenue and interest revenue.

**Expenses** are incurred **when a business receives goods and services**. Like revenues, payment **may or may not have been made**. Examples: salaries expense, utility expense and interest expense.

This is the key rule for the exercises: a revenue is recorded when the **sale** happens, not when the customer **pays**; an expense is recorded when the good or service is **received** (consumed), not when the firm **pays**. The difference between them and the cash movements is what creates receivables (sold but not collected) and payables (received but not paid) in the balance sheet.

## § 7 The income statement: extended format [Slides 9-11 ]

### 7.1 The format [Slide 9]

| Line | Operation |
|---|---|
| Sales revenue | |
| – Cost of goods sold | |
| **= Gross profit** | |
| – Operating expenses | |
| **= Income from operations** | |
| ± Non-operating items | |
| **= Income before taxes** | |
| – Income taxes | |
| **= Net income** | |

Each subtotal answers a different question: **gross profit** measures what is earned on the products themselves; **income from operations** adds the cost of running the business; **income before taxes** adds everything not related to the primary operations; **net income** is what is left for the owners after taxes.

### 7.2 Cost of goods sold and operating expenses [Slide 10]

- **Cost of goods sold** represents the expense a business incurred to **buy or make a product for resale**. Example on the slide: a book store buys a book for 25 USD and then sells it for 32 USD; the cost of goods sold is 25 USD (and the gross profit on that sale is $32 - 25 = 7$ USD).

- **Operating expenses** are the **usual expenses incurred in operating a business**. Accounts such as **salaries expense, utility expenses and depreciation expenses** are all shown in this section.

### 7.3 Non-operating items and income taxes [Slide 11]

- **Non-operating items** are revenues, expenses, gains and losses that **do not relate to the company's primary operations**. Accounts include **interest expense** and **gains and losses on the sale of equipment and investments**.
- **Income taxes** are computed by multiplying **income before taxes** by the **income tax rate**.

**Example on the slide:** income before taxes = 50,000 USD and income tax rate = 30%:

$$\text{Income taxes} = 50{,}000 \times 0.30 = 15{,}000 \text{ USD}, \qquad \text{Net income} = 50{,}000 - 15{,}000 = 35{,}000 \text{ USD}$$

## § 8 The balance sheet: purpose and basic equation [Slide 12 ]

The purpose of the balance sheet is to report the **financial position** of an accounting entity **at a particular point in time** (a photograph on one date, while the income statement is a film of a period).

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

## § 9 Assets, liabilities and equity [Slide 13 ]

- **Assets** are **economic resources owned** by a company. Examples: cash, accounts receivable, supplies, buildings and equipment.
- **Liabilities** are the company's **debt or obligations**. Examples: accounts payable, unearned revenues and bonds payable.
- **Equity** is the **residual balance**:

$$\text{Equity} = \text{Assets} - \text{Liabilities}$$

- Equity is commonly called **stockholders' equity** if the business is a corporation, as it represents the **financing provided by the stockholders** along with the **earnings from the business not paid out as dividends (retained earnings)**.

The residual definition is why the balance sheet always balances: equity is whatever makes the two sides equal. If the firm earns a profit and keeps it, both assets (more cash or receivables) and equity (more retained earnings) grow by the same amount.

## § 10 Current and non-current assets [Slides 14-15 ]

$$\text{Current assets} + \text{Non-current assets} = \text{Total assets}$$

- **Current assets** are assets that will be **used or turned into cash within one year**. Examples: cash, accounts receivable, inventory, short-term investments, supplies and prepaids [Slide 14].
- **Non-current assets** are the **remainder of the assets**. They include long-term investments, land, building, equipment and patents [Slide 15].

## § 11 Current and long-term liabilities [Slide 16 ]

$$\text{Current liabilities} + \text{Long-term liabilities} = \text{Total liabilities}$$

- **Current liabilities** are obligations that will be **paid in cash (or other services) or satisfied within the coming year**.
- **Long-term liabilities** are obligations that will **not be paid or satisfied within the year**.

The classification looks at **when** the obligation is due, measured from the balance-sheet date. A loan that is repaid in instalments is therefore split: the instalments due in the next twelve months are current, the rest are long-term.

## § 12 Stockholders' equity [Slide 17 ]

$$\text{Contributed capital} + \text{Retained earnings} = \text{Total stockholders' equity}$$

**Contributed capital** is what the stockholders put into the company; **retained earnings** are the earnings not paid out as dividends ([Slide 13]).

## § 13 The complete balance sheet [Slide 18 ]

| **Assets** | **Liabilities and stockholders' equity** |
|---|---|
| Current assets | Current liabilities |
| + Non-current assets | + Long-term liabilities |
| | + Stockholders' equity |
| **= Total assets** | **= Total liabilities and stockholders' equity** |

$$\text{Current assets} + \text{Non-current assets} = \text{Current liabilities} + \text{Long-term liabilities} + \text{Stockholders' equity}$$

## § 14 Worked example on the slides [Slide 19 ]

The slide gives a list of accounts, colour-coded as **assets, liabilities, equity, revenues and expenses**, and an income tax rate of 30%. It does not print the solution; the statements below are built from its data with the rules of Slides 9-18.

**Step 1: classify each account.**

| Account | Amount | Class | Where it goes |
|---|---|---|---|
| Cash | 5,000 | Asset | Current assets [Slide 14] |
| Accounts receivable | 10,000 | Asset | Current assets [Slide 14] |
| Inventories | 45,000 | Asset | Current assets [Slide 14] |
| Supplies | 4,000 | Asset | Current assets [Slide 14] |
| Buildings | 65,000 | Asset | Non-current assets [Slide 15] |
| Accounts payable | 12,000 | Liability | Current liabilities [Slide 16] |
| Bonds payable | 40,000 | Liability | Long-term liabilities (bonds finance long-term investments [Slide 29]) |
| Common stock | 45,000 | Equity | Contributed capital [Slide 17] |
| Additional paid-in capital | 20,000 | Equity | Contributed capital [Slide 17] |
| Retained earnings (beginning balance) | 5,000 | Equity | Retained earnings [Slide 17] |
| Sales | 100,000 | Revenue | Sales revenue [Slide 9] |
| Cost of goods sold | 58,000 | Expense | Cost of goods sold [Slide 10] |
| Utility expense | 8,000 | Expense | Operating expenses [Slide 10] |
| Supplies expense | 3,000 | Expense | Operating expenses [Slide 10] |
| Salaries expense | 16,000 | Expense | Operating expenses [Slide 10] |
| Interest expense | 5,000 | Expense | Non-operating items [Slide 11] |

Note the difference between **supplies** (4,000, an asset: supplies still in stock) and **supplies expense** (3,000, an expense: supplies consumed in the period). The same good is an asset while it is owned and becomes an expense when it is used.

**Step 2: income statement** (format of [Slide 9]).

| Line | Computation | Amount |
|---|---|---|
| Sales revenue | | 100,000 |
| – Cost of goods sold | | 58,000 |
| **= Gross profit** | $100{,}000 - 58{,}000$ | **42,000** |
| – Operating expenses | $8{,}000 + 3{,}000 + 16{,}000$ | 27,000 |
| **= Income from operations** | $42{,}000 - 27{,}000$ | **15,000** |
| – Non-operating items (interest expense) | | 5,000 |
| **= Income before taxes** | $15{,}000 - 5{,}000$ | **10,000** |
| – Income taxes (30%) | $10{,}000 \times 0.30$ | 3,000 |
| **= Net income** | $10{,}000 - 3{,}000$ | **7,000** |

**Step 3: retained earnings at the end of the period.** No dividends are given, so the whole net income is retained:

$$\text{Retained earnings}_{\text{end}} = 5{,}000 + 7{,}000 = 12{,}000$$

**Step 4: balance sheet** (format of [Slide 18]).

| Assets | Amount | Liabilities and stockholders' equity | Amount |
|---|---|---|---|
| Cash | 5,000 | Accounts payable | 12,000 |
| Accounts receivable | 10,000 | **Current liabilities** | **12,000** |
| Inventories | 45,000 | Bonds payable | 40,000 |
| Supplies | 4,000 | **Long-term liabilities** | **40,000** |
| **Current assets** | **64,000** | Common stock | 45,000 |
| Buildings | 65,000 | Additional paid-in capital | 20,000 |
| **Non-current assets** | **65,000** | Retained earnings | 12,000 |
| | | **Stockholders' equity** | **77,000** |
| **Total assets** | **129,000** | **Total liabilities and equity** | **129,000** |

**Check:** $64{,}000 + 65{,}000 = 129{,}000 = 12{,}000 + 40{,}000 + 77{,}000$. The balance closes with no tax liability, which means that the cash figure of the slide is already the cash after the 3,000 of income taxes were paid: had the taxes been unpaid, a liability of 3,000 would appear on the right and the two sides would differ by exactly 3,000.

# Part III — The financial structure of the company

## § 15 Financing decisions and the capital structure [Slide 20 ]

Financing decisions require the **analysis of the different financial alternatives**.

- **Business financing** consists in the **collection of the financial resources needed to develop the company's activity**.
- The **combination of all the possible financial sources** is known as the **CAPITAL STRUCTURE** or **FINANCIAL STRUCTURE**.

## § 16 The map of the financial structure [Slides 21-22 ]

The deck shows the same map twice, first by **legal nature** [Slide 21] and then by **origin** [Slide 22], and repeats it before each instrument with the one being explained highlighted (Slides 24, 27, 28, 30, 32 and 34).

| By nature [Slide 21] | By origin [Slide 22] | Instruments |
|---|---|---|
| **Equity** | **Company resources** | Internal equity: self-financing |
| | | External equity: equity capital and shares |
| **Liabilities** | **External resources** | Long term: bonds, long-term loans and bank credits, leasing |
| | | Short term: accounts payable, trade discount, short-term loans and bank credits, factoring |

The two classifications cut the same list in the same place: equity is the company's own resources (its owners' money, whether contributed or retained), and liabilities are external resources (money owed to third parties).

## § 17 Internal equity: self-financing [Slides 23-24 ]

`SELF-FINANCING [SLIDE 23]`

The **retained earnings** (resources generated within the company in its normal activity) constitute the **self-financing** of the company. These are the resources generated by the company.

**Profits** can be:

- **retained in the company** = **self-financing**;
- **spread** (distributed) = **dividends**.

Two ratios describe the split:

$$\text{Pay-out} = \frac{\text{Dividends}}{\text{Profits}}, \qquad \text{Self-financing (\%)} = \frac{\text{Retained earnings}}{\text{Profits}}$$

Since every euro of profit is either distributed or retained, $\text{Dividends} + \text{Retained earnings} = \text{Profits}$, and dividing by the profits:

$$\text{Pay-out} + \text{Self-financing (\%)} = 1$$

**Example on the slide:** profits = 100,000 €, dividends = 75,000 €, retained earnings = 25,000 €:

$$\text{Pay-out} = \frac{75{,}000}{100{,}000} = 0.75 = 75\%, \qquad \text{Self-financing} = \frac{25{,}000}{100{,}000} = 0.25 = 25\%$$

[Slide 24] repeats the map of the financial structure, now highlighting external equity.

# § 18 External equity [Slides 25-26 ]

## 18.1 Equity capital [Slide 25]

When a business is created, owners put some funding into the business to finance assets.

> EQUITY CAPITAL [SLIDE 25]
> **Equity capital** are the funds raised from owners **at the start of a business (initial capital)** and **further rights-issues (issuing new shares)**, used to finance the investments of the company.

Two features distinguish it from debt:

- it is **not to be paid back** to the investor at any expiring date;
- a share of capital can **only** be converted into cash **by sale or transmission** (the investor recovers the money by selling the share to someone else, not by asking the company for it).

## 18.2 Equity securities: shares [Slide 26]

- A **share of stock** is one of a **finite number of equal portions in the capital** of a company, entitling the owner to a proportion of the **distributed, non-reinvested profits, known as dividends**, and to a portion of the **value of the company in case of liquidation**.
- Shares are **equity securities**.
- A share has **different values**: **nominal**, **accounting (book)**, **intrinsic** and **market** value.
- New shares are issued in a **rights-issue (capital increase)**.
- **Payment of dividends is not compulsory.**

[Slides 27-28] repeat the map, now highlighting the external resources and, within them, bonds.

# § 19 Long-term external resources: bonds [Slide 29 ]

> BOND [SLIDE 29]
> A **bond** is a **debt security**, in which the authorized issuer owes the holders a debt and is obliged to **repay the principal and interest (the coupon)** at a later date, termed **maturity**.

It is a **loan in the form of a security**:

| Bond term | Loan term |
|---|---|
| Issuer | Borrower |
| Bond holder | Lender |
| Coupon | Interest |

Bonds enable the issuer to **finance long-term investments with external funds**.

[Slide 30] repeats the map, highlighting long-term loans and bank credits.

## § 20 Long-term loans and bank credits [Slide 31 ]

- **Long-term loans:** the company receives an **amount of money, which needs to be paid back at a certain time**. This service is provided at a cost: the **debt interest**.
- **Bank credits:** the financial institution makes available a **maximum amount of money** to the customer. The customer will then borrow **any amount of money under the limit**, paying **interests only for the amount borrowed** (**interest rates are higher**).

The difference: with a loan the whole amount is received and pays interest from the first day; with a credit the firm draws only what it needs and pays interest only on that, at a higher rate.

[Slide 32] repeats the map, highlighting leasing.

## § 21 Leasing [Slide 33 ]

LEASING [SLIDE 33]

**Leasing** is a process by which a firm can obtain **the use of a certain fixed asset** for which it must pay a **series of contractual, periodic payments**. The **lessee** (receiver of the services or the assets under the lease contract) has the **right to buy the asset at the end** of the agreed period.

**Types: finance leasing** and **operating leasing** (the slide names the two types and does not develop them further).

[Slide 34] repeats the map, highlighting the short-term external resources.

## § 22 Short-term external resources [Slide 35 ]

- **Accounts payable:** financing from suppliers (**trade credit**) arises when payment of goods is done **some time after their reception**. Suppliers offer the company the possibility to **defer the payment**.
- **Bank financing:**
- **loans and credits**, with the same characteristics as the long-term ones;
- **trade discounts**: the financial institution makes a note (bill of exchange, promissory note, ...) **available to the customer before its expiring date**, having previously **deducted some expenses** (interest rate, commission, ...).
- **Factoring:** the company **sells to a factoring company a credit from a customer, before it expires**. The factoring company is in charge of its reimbursement, **assuming the risk of insolvency**.

In the exercises, accounts payable are the most frequent short-term resource: every purchase "paid at 60 or 90 days" is trade credit, and while it is unpaid it appears as a current liability.

## § 23 Measuring the financial structure: leverage [Slide 36 ]

The **financial structure** is the **combination of the different financial resources**. With $E$ = equity and $L$ = liabilities:

$$\%\,\text{Equity} = \frac{E}{E+L}, \qquad \%\,\text{Liabilities} = \frac{L}{E+L}, \qquad \text{Debt-to-equity ratio} = \frac{L}{E}$$

The **debt-to-equity ratio** is used to evaluate a company's **financial leverage**. It measures the degree to which a company is financing its operations **through debt versus wholly-owned funds**. It reflects the **ability of shareholder equity to cover all outstanding debts** in the event of a business downturn. **Higher leverage ratios tend to indicate a company or stock with higher risk to shareholders.**

**Example on the slide:** total liabilities + equity = 1,000 M€, equity = 700 M€, liabilities = 300 M€.

$$\%\,\text{Equity} = \frac{700}{1{,}000} = 70\%, \qquad \%\,\text{Liabilities} = \frac{300}{1{,}000} = 30\%, \qquad \frac{L}{E} = \frac{300}{700} = 0.43$$

So the company has 0.43 € of debt for every euro of equity. The first two ratios always add up to one, because $\frac{E}{E+L} + \frac{L}{E+L} = \frac{E+L}{E+L} = 1$.

## § 24 The cost of the financial resources: WACC [Slide 37 ]

**All the financial resources have a cost:**

- **implicit cost (opportunity cost): equity**. Nothing is paid by contract, but the owners give up what they could earn elsewhere with that money;
- **explicit cost: liabilities**. The interest is fixed by contract.

CAPITAL COST [SLIDE 37]

The **capital cost** is the **weighted average cost** associated to all the financial resources obtained by the company ($K$):

$$K_{WACC} = K_{EQ} \cdot \frac{E}{E+L} + K_L \cdot \frac{L}{E+L}$$

where $K_{EQ}$ is the cost of equity, $K_L$ the cost of liabilities, and the weights are the two percentages of [Slide 36]. With the structure of the slide-36 example (70% equity, 30% liabilities), the formula reads

$$K_{WACC} = 0.70\,K_{EQ} + 0.30\,K_L$$

so the cost of capital is always between the cheaper and the dearer of the two sources, closer to the one with the larger weight. This closes the argument of [Slide 5]: the investments (assets) create value only if their profitability is greater than this cost of the funds that finance them.

Solved exercises for this topic (sheet exercises 1-6 and the four additional exam-type problems) →

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Study notes written from the official course material (Topic 3 Financial Management (I).pdf; T3_Financial Statements_EXERCISES.pdf). Online version: subjects/business-management/teoria/topic-3.html